



TASHKENT STATE  
UNIVERSITY OF ECONOMICS

# NEW2AN 2022

Next Generation Wired/Wireless Networks and Systems



Springer

Lecture Notes in  
Computer Sciences (LNCS)

SCIENTIFIC PARTNERS







**THE 22ND INTERNATIONAL CONFERENCE ON NEXT GENERATION  
WIRED/WIRELESS NETWORKS AND SYSTEMS  
(NEW2AN 2022)**

Hosted as a Hybrid of On-Site and Virtual Format

[http:// new2an.info](http://new2an.info)

Tashkent, Uzbekistan

15th-16th December, 2022

Future mobile networks will evolve out of existing wired and wireless networks of different nature by means of convergence, fusion and mutual penetration; next-generation multimedia services in turn pose new requirements and challenges on treatment of traffic within communication systems - a certain degree of multilayer co-operation has to be achieved. This forms a framework on harmonization of results obtained separately in different areas of network research like network performance evaluation, architecture and protocol design, and low layer network design. The conference goal is in the identification, investigation and integration of new algorithms, approaches, architectures, methods and mechanisms to enable proper and efficient operation of a next-generation IP-based wireless network. Therefore, wireless networks and systems and their interaction with wired networks and systems shall be widely examined and addressed throughout the conference.

The history of NEW2AN dates back to 1993, today it is an established conference featured by proceedings published by Springer LNCS and distinguished keynote speakers.

**CONFERENCE PROGRAMME**  
**THURSDAY, DECEMBER 15, 2022**

|                | TIME          | ACTIVITY                               | PRESENTER(S)                                                                                                                    |
|----------------|---------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>PLENARY</b> | 09:00 – 10:00 | <i>Registration (Madaniyat Saroyi)</i> |                                                                                                                                 |
|                | 10:00 – 10:05 | Opening Address                        | <b>Prof. Ahmed Mohamed Aziz Ismail</b><br>Head of the Business Administration<br>Department, TSUE & IMC Krems<br>University     |
|                | 10:05 – 10:15 | Welcoming speech                       | <b>Prof. Kongratbay Sharipov</b><br>Rector of the Tashkent State University of<br>Economics, Uzbekistan                         |
|                | 10:15 – 10:25 | Welcoming address                      | <b>Mr. Yousef Saleh Algahrah Al-Otaibi</b><br>Ambassador Extraordinary and<br>Plenipotentiary of the Kingdom of Saudi<br>Arabia |
|                | 10:25 – 10:35 | Welcoming address                      | <b>Ibrokhim Abdurakhmonov</b><br>Minister of Innovative Development of the<br>Republic of Uzbekistan                            |
|                | 10:35 – 10:45 | Welcoming address                      | <b>Prof. Mansur Eshov</b><br>Vice-Rector for Academic Affairs of the<br>Tashkent State Economic University                      |

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|--------------|---------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | 10:45 – 10:55 | Welcoming address                                                                     | <b>Prof. Gulnora Abdurakhmanova</b><br>Vice-Rector for Research and Innovations,<br>Tashkent State Economic University                                                                      |
|              | 10:55 – 11:05 | Welcoming address and highlights of ICFNDS-2022                                       | <b>Prof. Mohammad Hammoudeh</b><br>Saudi Aramco Chair Professor of Cyber Security in the Information & Computer Science Department at King Fahd University of Petroleum & Minerals (KFUPM). |
|              | 11:05-11:15   | Welcoming address and highlights of NEW2AN-2022                                       | <b>Prof. Evgeny Kucheryavy</b><br>Faculty of Information Technology and Communication Sciences, Tampere University, Finland                                                                 |
|              | 11:15-11:30   | <i>Memento Award</i>                                                                  |                                                                                                                                                                                             |
|              | <b>BREAK</b>  | 11:30 – 11:45                                                                         | <b>Coffee-break</b>                                                                                                                                                                         |
|              | 11:45 – 12:45 | <b>Tech Talk</b>                                                                      | <b>Key note Speakers</b>                                                                                                                                                                    |
|              | 11:45 – 12:00 | Effective cyber risk management through enhancing organizational capacity and process | <b>Ninomiya Hideki</b><br>Division of Environmental Engineering, Hokkaido University, Japan                                                                                                 |
|              | 12:00 – 12:15 | Reinforcement Learning to build efficient swarm intelligence algorithm                | <b>Prof. Aydin Mehmet Emin,</b><br>Department of the Computer Science, University of the West of England, UK                                                                                |
|              | 12:15 – 12:30 | The role of IR4.0 in digital transformation.                                          | <b>Prof. Mohamed Helmy Abd Wahab</b><br>University Tun Hussien Onn Malaysia, Malaysia                                                                                                       |
|              | 12:30 – 12:45 | Cyber Iot, future and challenges                                                      | <b>Prof. Abdelkader Laouid</b><br>University of El Oued, AGARIA                                                                                                                             |
|              | 12:45 – 13:00 | Twitter in Uzbekistan: Social Media and Political Stability                           | <b>Prof. Mladen Mrdalj</b><br>Webster University, Tashkent                                                                                                                                  |
| <b>BREAK</b> | 13:00 – 14:00 | Lunch (Madaniyat saroyi, "Sokin" )                                                    |                                                                                                                                                                                             |

| Decentral Parallel Sessions                                                                                                                 |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <b>DAY 1: December 15</b><br><b>SESSION 1 Communication topics I</b><br><b>Starts at 14:00</b><br><b>Moderator: Prof. Evgeny Kucheryavy</b> |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>SESSION 1<br/>ROOM-111</b>                                                                                                               | 14:00 – 14:10 | Navruzov K., Fayziev R.A., Mirzoev A.A., Sharipova Sh. “Tangential shear stress in an oscillatory flow of a viscoelastic fluid in a flat channel”, Urganch State University, Tashkent State University of Economics, Institute of Mechanics and Seismic stability of structures, Uzbekistan                                                                                                                                                                      |
|                                                                                                                                             | 14:10 – 14:20 | Murodil Erkinjon ugli Madaliev, Rabim Alikulovich Fayziev, Eshmurod Sattarovich Buriev, Akmal Ahadovich Mirzoev “Comparison of Finite Difference Schemes of Different Orders of Accuracy for the Burgers Wave Equation Problem”, Fergana Polytechnic Institute, Tashkent Institute of Architecture and Civil Engineering, Tashkent State University of Economics, Uzbekistan                                                                                     |
|                                                                                                                                             | 14:20 – 14:30 | Ismatulla Qo’shaevich Khujaev, Rabim Alikulovich Fayziev, Muzaffar Muhiddinovich Hamdamov “Numerical solution of the combustion process using the computer package ANSYS fluent”, Institute of Mechanics and Seismic Stability of Structures of the Academy of Sciences of the Republic of Uzbekistan and Tashkent State University of Economics, Uzbekistan                                                                                                     |
|                                                                                                                                             | 14:30 – 14:40 | Abidov Abdujabbar Abdukhamidovich “Simulation Modeling of Reliability of Packet Switching Unit”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                              |
|                                                                                                                                             | 14:40 – 14:50 | Abdujabbor Abidov, Dilmurod Mirzaaxmedov, Dilmurod Rasulev “Analytical Model for Assessing the Reliability of the Functioning of the Adaptive Switching Node”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                |
|                                                                                                                                             | 14:50 – 15:00 | Dilafruz Nurjabova, Qulmamatova Sayyora, Pardayeva Gulmira “Artificial intelligence software architecture in the field of cardiology and application in the cardio vessel project using CJM and customer development methods”, Tashkent University of Information Technologies, Tashkent State University of Economics, Uzbekistan                                                                                                                               |
|                                                                                                                                             | 15:00 – 15:10 | Dilafruz Nurjabova, Qulmatova Sayyora, Pardayeva Gulmira “Using discretization and numerical methods of problem 1d-3d-1d model for blood vessel walls with Navier-Stokes”, Tashkent University of Information Technologies, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                   |
|                                                                                                                                             | 15:10 – 15:20 | Zafar Mamatkulovich Malikov, Farrukh Kholiyorovich Nazarov, Bekzod Sultonmurodovich Toshpulov, Bekhromjon Alisherovich Abdurakhmonov “Numerical simulation of a flow in a two-dimensional channel on the basis of a two-liquid turbulence model”, Institute of Mechanics and Seismic Stability of Structures of the Academy of Sciences of the Republic of Uzbekistan, Tashkent Pharmaceutical Institute and Tashkent State University of Economics, Uzbekistan  |
|                                                                                                                                             | 15:20 – 15:30 | Rakhimberdiev Kuvonchbek, Toshpulov Bekzod, Kamalov R. Abdullaev O.K., Khojaniyazova S. “Application fuzzy neural network methods to detect cryptoattacks on financial information systems based on blockchain technology”, National University of Uzbekistan, Tashkent State University of Economics, Karakalpak State University, Tashkent University of Information technologies, Nukus branch of Tashkent University of Information technologies, Uzbekistan |

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|  | 15:30 – 15:40 | Dilorom Ergasheva, Akram Ishnazarov “Method authentication of objects information communication systems”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                            |
|  | 15:40 – 15:50 | Walid Osamy, Ahmed M. Khedr, Ahmed Salim, Ahmed Aziz “TEDCTSSA: Trust Enabled Data Collection Technique based Sparrow Search Algorithm for WSN-based Applications”, Benha University, Egypt, University of Sharjah, UAE, Qassim University, Saudi Arabia, Zagazig University, Egypt, Tashkent State University of Economics, Uzbekistan |
|  | 15:50 – 16:00 | Reham R. Mostafa, Ahmed Aziz, Ahmed M. Khedr “ISTOA: An improved Sooty Tern Optimization Algorithm for multilevel threshold image segmentation”, Mansoura University, Egypt, Tashkent State University of Economics, Uzbekistan, University of Sharjah, UAE                                                                             |

| <b>SESSION 2 Digitalization of economies</b><br><b>Starts at 14:00</b><br><b>Moderator: Prof. Mohamed Helmy Abd Wahab</b> |               |                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <b>SESSION<br/>2<br/>ROOM-<br/>118</b>                                                                                    | 14:00 – 14:10 | Shadibekova Dildor, Ismoilov Narimonjon “Implementing digital transformation in the logistics system of Uzbekistan”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                        |
|                                                                                                                           | 14:10 – 14:20 | Muzaffar Muhiddinovich Hamdamov, Akram Ismoilovich Ishnazarov, Khusniddin, Abdujalilovich Mamadaliev “Numerical modeling of vertical axis wind turbines using ANSYS FLUENT software”, Institute of Mechanics and Seismic Stability of Structures of the Academy of Sciences of the Republic of Uzbekistan, Tashkent University of Information Technologies, Tashkent State University of Economics, Uzbekistan |
|                                                                                                                           | 14:20 – 14:30 | Urmanov Bahromjon, Shin Hoyoung, Abdullaev Munis, Suleimanov Farrukh, Uktamova Durdon, Bakhodirova Durdon, Norboyeva Nafisa ““I’ll wait 4 ur answr!” A Study on Modern Style of Cyber-Writing and User Reactions”, Tashkent State University of Economics, Uzbekistan and Yeungnam University, South Korea                                                                                                     |
|                                                                                                                           | 14:30 – 14:40 | Gayrat Ishankhodjayev, Murodjon Sultanov, Rano Parpiyeva and Nafisa Norboyeva “Improvement of information support in intelligent information energy systems”, Institute of Energy Problems and Tashkent State University of Economics, Uzbekistan                                                                                                                                                              |
|                                                                                                                           | 14:40 – 14:50 | Fayzieva Muyassarzoda “The Assessment of the Effectiveness of the Development of Digital Technologies in Commercial Banks in Uzbekistan”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                   |
|                                                                                                                           | 14:50 – 15:00 | Fayzieva Muyassarzoda, Mirzaliev Sanjar, Yuldashev Maksudjon, Bakiyeva Iroda, Kamilova Iroda, Oqmullayev Ravshan “The Study of the Impact of the Digital Economy on the Growth of E-Government Services in Uzbekistan”, Tashkent State University of Economics, Uzbekistan                                                                                                                                     |
|                                                                                                                           | 15:00 – 15:10 | Rashid Nasimov, Nigorakhon Nasimova, Karimov Botirjon, Munis Abdullayev “Deep learning algorithm for classifying dilated cardiomyopathy and hypertrophy cardiomyopathy in transport workers”, Tashkent State University of Economics, Tashkent University of Information Technologies, Uzbekistan                                                                                                              |
|                                                                                                                           | 15:10 – 15:20 | Faridakhon Abdukarimovna Khamidova, Shakhzod Sherzodovich Saydullaev “eCommerce benchmarking: theoretical background, variety of types, and application of competitive-integration benchmarking”, Tashkent Financial Institute, Tashkent State University of Economics, Uzbekistan                                                                                                                             |
|                                                                                                                           | 15:20 – 15:30 | Shokirov Mirkamol, Eshov Mansur “Cryptocurrencies as the money of the future”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                              |
|                                                                                                                           | 15:30 – 15:40 | Ahmed Aziz, Ahmed M. Khedr, Ahmed Salim and Walid Osamy “A Data Security Technique Combining Asymmetric Cryptography and Compressive Sensing for IoT Enabled Wireless Sensor Networks”, Benha University, Egypt, University of Sharjah, UAE, Qassim University, Saudi Arabia, Zagazig University, Egypt, Tashkent State University of Economics, Uzbekistan                                                    |
|                                                                                                                           | 15:40 – 15:50 | Ahmed Aziz, Maha Ibrahim “Energy Efficient and Secure Scheme based Compressive Sensing method for Internet of Vehicles”, Benha University, Egypt, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                           |

**SESSION 3 Digitalization of financial sphere****Starts at 14:00****Moderator: Prof. Muhammad Balbaa**

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| <b>SESSION<br/>3<br/><br/>ROOM<br/>115</b> | 14:00 – 14:10 | Irmatova Aziza Bakhramovna , Akbarova Malika Ilhomovna “Impact of digital technologies on women's employment”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                        |
|                                            | 14:10 – 14:20 | Oksana Bondarskaya, Jamshid Tukhtabaev , Rano Akramova , Kholida Saidrasulova , Uchkun Shirinov, Zulaykho Kurbanova “The impact of digitalisation on the safe development of individuals in society”, Tambov State Technical University, Tashkent State University of Economics, Tashkent Chemical - Technological Institute, University of Geological Sciences, Samarkand Institute of Economics and Service, Termez branch of the Tashkent Medical Academy, Uzbekistan |
|                                            | 14:20 – 14:30 | Yuldashev Abdulkhakim “Econometric Evaluation of the Efficiency of the Management of the Enterprise through the Supply of Raw Materials in Oil Enterprises in the Conditions of the Digital Economy”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                 |
|                                            | 14:30 – 14:40 | Lochinbek Amirov, Nuriddin Avazov “The impact of digital infrastructure, foreign direct investment and trade openness on economic growth: In the case of Uzbekistan”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                 |
|                                            | 14:40 – 14:50 | Bunyod Usmonov “The impact of the financial ratios on the financial performance. A case of Chevron Corporation (CVX)”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                |
|                                            | 14:50 – 15:00 | Gaipov Jasur Bakhrom ugli “The impact of the digitalisation of payment systems on the profitability of commercial banks”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                             |
|                                            | 15:00 – 15:10 | Qosimova Dilorom Sabirovna and Nutfulloev Tolib G’olib o’g’li “The main aspects and benefits of digital transformation of business entities”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                         |
|                                            | 15:10 – 15:20 | Choriev Fazliddin Ishquvvatovich “The influence of the capital structure of state enterprises on the profitability of the enterprise”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                |

| <b>SESSION 4 Other topics</b><br><b>Starts at 14:00</b><br><b>Moderator: Dr. Sherzod Dadaboyev</b> |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>SESSION<br/>4<br/>ROOM-<br/>116</b>                                                             | 14:00 – 14:10 | Shufeng Cong , Lee Chin, Piratdin Allayarov “Exploring the development of China's digital trade in the context of the domestic and international double cycle”, Universiti Putra Malaysia and Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                |
|                                                                                                    | 14:10 – 14:20 | Abdurashidova Nigora, Valiyeva Aziza, Tuychieva Vasila, Mumtozbegin Kholmatova, Minarova Murshida Khojimuratovna “A systematic mapping study of using the cutting-edge technologies in marketing: the state of the art of four key new-age technologies”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                    |
|                                                                                                    | 14:20 – 14:30 | Bahodir Zaripov, Feruza Zakirova, Sanjar Mirzaliev “Social Media Marketing for Educational Purposes: Goals, Objectives and Content of the Training Course”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                                  |
|                                                                                                    | 14:30 – 14:40 | Abdulaziz Aliev , Djavlonbek Kadirov “Digital Marketing and Smart Technology Marketing Systems as the future of metaverse”, Tashkent State University of Economics, Uzbekistan and Victoria University of Wellington, New Zeland                                                                                                                                                                                                                                                                                                |
|                                                                                                    | 14:40 – 14:50 | Abdurashidova Marina Sagatovna, Muhammad Eid BALBAA “The impact of the digital economy on the development of higher education”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                                                              |
|                                                                                                    | 14:50 – 15:00 | Ochilova Gulnoza Odilovna , Musaxonova Gulnora Mavlyanovna , Miraliyeva Dilafruz Toxirovna , Akbarova Sayyora Shuxratovna , Karimova Feruza Xamidullayevna “What is the state-of-the-art contribution of the higher education system to the digital economy: a systematic mapping study on changes and challenges”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                          |
|                                                                                                    | 15:00 – 15:10 | Yuldasheva Sharapat, Mirtursunova Yulduz , Miralieva Dilafruz , Bakieva Hilola , Shodmonkulova Dilorom “Innovating primary education of promoting students' language competencies through mobile assisted language learning approach: Selection framework of innovative digital technologies”, Tashkent State Pedagogical University and Tashkent State University of Economics, Uzbekistan                                                                                                                                     |
|                                                                                                    | 15:10 – 15:20 | Jamshid Sharafetdinovich Tukhtabaev, Gulnoza Toxirovna Samiyeva, Kushbakov Abduxamit Norjigitovich, Aziza Abdusalomovna Goziyeva, Razakova Barno Sayfiyevna, Aktamov Olimjon Abdugani ugli “Econometric assessment of the dynamics of development of the export authority of small business and private business subjects in the conditions of the digital economy”, Tashkent State University of Economics, Karshi Engineering and Economic Institute, Tashkent Medical Academy, National University of Uzbekistan, Uzbekistan |
|                                                                                                    | 15:20 – 15:30 | “The significance of the Internet of Things for ensuring the smooth operation of network functions in fintech”, Olim Astanakulov, Muhammad Eid Balbaa, International Islamic Academy of Uzbekistan and Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                       |
|                                                                                                    | 15:30 – 15:40 | Aziza Usmanova “Impact of E-government on Poverty Rate: a Cross-Country Empirical Assessment”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                    | 15:40 – 15:50 | Aziza Usmanova “An empirical investigation of the relationship between e-government development and multidimensional poverty”, Tashkent State University of Economics, Uzbekistan                                                                                                                                                                                                                                                                                                                                               |



**DAY 2: December 16**  
**SESSION 5 Communication topics II (online)**  
**Starts at 14:00**

*Moderator: Prof. Ammar Muthanna*

*Zoom link: [https://us06web.zoom.us/j/4291808713?pwd=ia0ZRzB9UM0iMDYQCzF2rl\\_2ar8](https://us06web.zoom.us/j/4291808713?pwd=ia0ZRzB9UM0iMDYQCzF2rl_2ar8)*

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| <b>SESSION 5</b> | 14:00 – 14:10 | Manfred Sneys-Snepe, Dmitry Namiot “On Digital Twin Software and Cyber Threats”, Lomonosov Moscow State University, Ventspils University of Applied Sciences, Latvia                                                                                                                                                                                                                                                               |
|                  | 14:10 – 14:20 | Dmitry Namiot, Manfred Sneys-Snepe, Vladimir Sukhomlin “On local services based on non-standard Wi-Fi Direct usage model”, Lomonosov Moscow State University, Ventspils University of Applied Sciences, Latvia                                                                                                                                                                                                                     |
|                  | 14:20 – 14:30 | Alexander Pastukh , Valeriy Tikhvinskiy , Evgeniy Devyatkin “Compatibility analysis between 5G NR and ultra-wideband devices in the 6425-7125 MHz frequency band”, Radio Research and Development Institute (NIIR), International Information Technologies University (IITU) Almaty, Kazakhstan                                                                                                                                    |
|                  | 14:30 – 14:40 | Valery Tikhvinskiy , Vitaly Urodlivchenko , Victor Koval , Alexander Frolov , Vladimir Ketat , Vadim Belyavskiy “6 GHz band sharing study for FWA base stations and GEO satellite receivers”, FSUE Radio Research & Development Institute (NIIR), Geyser-Telecom Ltd., International Information Technologies University (IITU) Almaty, Kazakhstan                                                                                 |
|                  | 14:40 – 14:50 | Amjad Ali “Federated Learning Strategies Over Wireless Channels”, Higher School of Economics University                                                                                                                                                                                                                                                                                                                            |
|                  | 14:50 – 15:00 | Sreejith Vidhyadharan, Paul Snyder, Mayank Anchlia, and Pulkil Agrawal “Data Routing in UAV Networks with Multiple Data Sources using Steiner Tree”, University of North Dakota, USA, BITS-Pilani KK Birla Goa Campus, India                                                                                                                                                                                                       |
|                  | 15:00 – 15:10 | Kirill Kozorez, Andrey Rashich “Reduced complexity distributed arithmetic architecture for FIR filters”, Peter the Great St.Petersburg Polytechnic University, St.Petersburg                                                                                                                                                                                                                                                       |
|                  | 15:10 – 15:20 | Truong Duy Dinh, Duc Tran Le, Khanh Quoc Dang, Vladimir Vishnevsky, Ruslan Kirichek “Blockchain-driven Hybrid Model for IoT Authentication”, V. A. Trapeznikov Institute of Control Sciences, The Bonch-Bruевич Saint Petersburg State University of Telecommunications, University of Science and Technology – The University of Danang, Danang, Vietnam and Posts and Telecommunications Institute of Technology, Hanoi, Vietnam |
|                  | 15:20 – 15:30 | Jerzy Martyna “An Heuristic Approach for Mapping of Service Function Chains in Softwarized 5G Networks”, Jagiellonian University, Poland                                                                                                                                                                                                                                                                                           |
|                  | 15:30 – 15:40 | Bashir Aliyu, Evgeny Mokrov, and Konstantin Samouylov “Multi-threshold hysteresis-Based Congestion Control for UAV-based Detection Sensor Network”, Peoples’ Friendship University of Russia (RUDN University), Institute of Informatics Problems, Federal Research Center “Computer Science and Control”                                                                                                                          |
|                  | 15:40 – 15:50 | Rostislav Shaniiazov, Anton Sergeev “Analysis of the capacity gain of Probability Shaping QAM”, Higher School of Economics University                                                                                                                                                                                                                                                                                              |
|                  | 15:50 – 16:00 | Van Dai Pham, Vladimir Vishnevsky, and Ruslan Kirichek “LoRa Mesh Network for Image Transmission: An Experimental Study”, Bonch-Bruевич Saint-Petersburg State University of Telecommunications, St. Petersburg, V. A. Trapeznikov Institute of Control Sciences                                                                                                                                                                   |